# Bayesian Network Reconstruction of Interbank Exposures (Nepal)

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**Abstract**

text text

## 1 Paper 1 — Bayesian Network Reconstruction of Interbank Exposures (Nepal)

**Status:** design settled, not yet implemented. **Target venue:** ICLR / NeurIPS / ICAIF, 7 pages.

**One-line description:** Reconstruct the latent bilateral interbank exposure matrix $W_t$ for Nepali BFIs from NRB bulletin data, using an SBM-structured, covariate-informed generative model with a differentiable regulatory-constraint projection layer, amortized by a GNN/GRU inference network.

This supersedes the earlier version of `feature_model_mapping.md`, which had drifted from the actual design (it invented a generic "Bayesian-net structural-constraint" layer built from accounting identities — that was never part of the real design and is corrected below).

### 1.1 0. Problem framing and contribution to policy

**Problem framing (what "the network" means here).** We do not pre-specify the edges of the interbank network. Instead, following the network reconstruction literature (e.g., Squartini et al., 2018), we treat the bilateral exposure structure as latent and infer it from publicly observable aggregates — balance-sheet marginals and bank-level covariates — within a constrained statistical framework. Concretely: a node is a bank; a directed edge $i \to j$ is $W_{ij,t}$, the exposure of bank $i$ to bank $j$; an edge "exists" when $W_{ij,t} > 0$. Neither the edge set nor the weights are observed — only the row/column marginals and covariates are. This is a problem of statistical inference under partial observability, not analysis over a topology that is already known or self-reported. (Revisit/tighten this paragraph later — current form is a first pass.)

**Contribution to policy.**

- Zero New Reporting Burden: No central bank in a resource-constrained setting is going to mandate new bilateral exposure reporting purely for a research exercise — that is a costly, politically slow regulatory change. The contribution here is that this doesn't require that change: a usable,

continuously-updatable estimate of systemic interconnectedness is produced from data NRB (or any similarly-resourced central bank) already publishes every month, with zero new reporting burden. That is, by relying exclusively on aggregate publicly available data, the model requires zero new regulatory compliannce overhead.

- Unlocking an answerable Supervisory Question: At present, there is no way to ask *who is exposed to whom* or *which banks are structurally systemic* — not because the question doesn't matter, but because answering it was assumed to require data that isn't collected. **This method shows it doesn't.** It converts a previously unanswerable supervisory question into an answerable one, cheaply, and produces a reusable pipeline that other analyses can plug into. In summary, the *data barrier* can be bypassed computationally. The method hence turns the blind spot in systemic risk monitoring into generalized framework providing where the future policy framework implemention can and should be aimed at.
- Generalization: This also generalizes beyond Nepal: most emerging-market central banks publish the same kind of aggregate call-report data and face the identical constraint, so the method is a template for resource-constrained regulators broadly, not a Nepal-specific workaround.

### 1.2 1. Why this is its own paper

Separated from the causal/policy question (Paper 2) because the two are methodologically unrelated: this paper's contribution is a reconstruction method with no causal-identification content; Paper 2's contribution is a causal-identification design that *consumes* this paper's output as a fixed input. Bundling them would force a methods paper to carry a policy narrative it doesn't need, and vice versa.

### 1.3 2. Notation

| Symbol | Meaning |
|---|---|
| $i, j \in \{1, ..., N\}$ | individual banking and financial institutions |
| $t \in \{1, ..., T\}$ | periods (NRB bulletin months) |
| $X_{i,t} \in \mathbb{R}^d$ | observed node covariates for bank $i$ at time $t$ |
| $r_{i,t}, c_{i,t} \geq 0$ | observed interbank lending / borrowing marginals |
| $K_{i,t}$ | core capitalof the bank, that defines the the regulatory single-obligor cap) |
| $W_t \in \mathbb{R}_{\geq 0}^{N \times N}$ | The true **latent** bilateral interbank exposure matrix — never observed, the reconstruction target |
| $z_i \in \{1, ..., K\}$ | latent block/tier membership (e.g. core vs. periphery) |

## 1.4 3. Generative model

- Block membership: Banks are sorted into latent tiers (SBM priors) : $z_i \sim \text{Categorical}(\pi)$ — (Eq. 1).
- Block-pair intensity: $\Lambda \in \mathbb{R}_{\geq 0}^{K \times K}$ — the SBM core
- Fitness/degree-correction from covariates: bank baseline activity scales with covariates: $\phi_{i,t} = \exp(w^\top X_{i,t})$ — (Eq. 2)
- Bilateral Latent edge intensity: the exposure magnigute is a fucntion of tier interaction and invidivudal fitness.

$$\lambda_{ij,t} = \Lambda_{z_i z_j} \cdot \phi_{i,t}\phi_{j,t} \cdot \exp(h_{i,t}^\top h_{j,t}) \qquad \text{(Eq. 3)}$$

- Zero-inflated edge weight (most bank pairs have *no* exposure). The edges are modeled via hurle process - a Bernoulli gate determines connection existence, followed by a LogNormal distribution for positive exposure volumes:

$$\Pr(\tilde{W}_{ij,t} = 0) = \sigma(-\eta_{ij,t}), \qquad \tilde{W}_{ij,t} \mid \tilde{W}_{ij,t} > 0 \sim \text{LogNormal}(\log \lambda_{ij,t}, \sigma^2) \qquad \text{(Eq. 4)}$$

- Marginals are a **deterministic function** of $W_t$:

$$r_{i,t} = \sum_j W_{ij,t}, \qquad c_{j,t} = \sum_i W_{ij,t} \qquad \text{(Eq. 5)}$$

This means the marginal constraint belongs in the *architecture* (a hard projection), not the likelihood — see §5. Rows and columns sums mach observed micro-prudential totals $(r_{i,t}, c_{i,t})$, enforced strictly via the Sinkhorn projection layer rather than soft likelihood penalties.

## 1.5 4. Inference network (amortized, PyG GNN + GRU)

- Design Principle: The encoder processes covariates/history only — it cannot message-pass over $W_t$ itself, since that would be circular (the whole point is that $W_t$ is unobserved). This design principle is implemented because in the standard graph autoencodersv(GAEs), the adjacency matrix defined by bilateral or multilateral relationshhips is an input to encoder which in turn uses message-passing across *known connections* to learn the node embeddings. In the current context, the true interbank $(W_t)$ network is not known publicly - that is why the paper aims to *reconstruct* interbank network from publicly available data. Because the encoder has an input $W_t$, but the goal is to figure out the what $W_t$ is, then modeling is trapped into logical circularity which is why this principle is adopted in order answer before it becomes burden in downstream task.
- Temporal encoder: $h_{i,t} = \text{GRU}_\theta(X_{i,t}, h_{i,t-1})$ (Eq. 6), optionally refined by a GNN layer over an auxiliary *observed* graph if one exists (e.g. common-sector exposure — see §8).
- Amortized block-assignment posterior: $q_\psi(z_i \mid X_{i,1:t}) = \text{Categorical}(\text{softmax}(\text{MLP}_\psi(h_{i,t})))$ (Eq. 7)
- Pairwise decoder produces $(\eta_{ij,t}, \lambda_{ij,t})$ from $(h_{i,t}, h_{j,t}, z_i, z_j)$ — full amortized variational posterior

$q_\psi(W_t \mid X_{1:t}, r_t, c_t)$

- Reparameterized sampling: Generates log-normal reparam trick for the continuous weight, and Gumbel-softmax for the discrete existence gate.

### 1.6 5. Constraint-satisfaction layer (the technical core)

Differentiable **capacitated Sinkhorn / entropic-optimal transport projection**:

$$W_t = \text{Sinkhorn}_{\text{cap}}(\tilde{W}_t;\ r_t, c_t, \kappa K_t) \qquad \text{(Eq. 8)}$$

Enforces, exactly and differentiably:

- 1. row/column marginals match the observed $r_{i,t}, c_{i,t}$
- 2. the regulatory single-obligor cap $W_{ij,t} \le \kappa K_{i,t}$

References: Cuturi (2013) entropic OT; Mena et al. (2018) Gumbel-Sinkhorn. Flagged explicitly as "the one genuinely nontrivial implementation piece" in the design discussion — this is the part most likely to need real engineering effort, not the GNN/GRU scaffolding around it.

**Correction vs. the earlier `feature_model_mapping.md`:** there is no "Bayesian-net structural-constraint edge" tying `TOTAL Liabilities = TOTAL ASSETS` or `Net Profit` into the graph. That was never part of this design. The accounting identities in the raw NRB statements are a **data-quality cross-check** on the extractor (do the reported totals reconcile?), not a component of the generative model or the constraint layer.

### 1.7 6. Training objective

$$\mathcal{L} = -\mathbb{E}_q[\log p(W_t^{\text{true}})]_{\text{synthetic only}} + \text{KL}(q_\psi(z) \,\|\, p(z)) + \lambda_{\text{smooth}} \sum_t \|W_t - W_{t-1}\|_F^2 + \lambda_{\text{reg}} \|\Lambda, w\|^2$$

- **Synthetic Validation data** (known ground-truth $W_t$ from an SBM generator): the first term is a real supervised reconstruction loss — this is the main validation evidence. That, when training on synthetic data generated via the SBM (where the true networks $W_t^{true}$ are known), the first term acts as a suprvised reconstruction loss, which in turn serves as the primary validation of the model's accuracy.

- **Real data** (no $W_t$ is ever observed): When applied to a real world central bank data where $W_t$ is unobserved, the supervised reconstruction term drops out entirely, the first term drops out entirely. What's left is a regularized/MAP-style constrained fit, *not* likelihood-based estimation. That is, in a standard statiscal or ML models, the model looks into observed ground trught answers (eg $W_t^{true}$). The model is trained by calculating the likelihood of model's guesses against the actual answers followed by parameter optimization to maximize the likelihood. However in this case, the true $W_t^{true}$ does not exist in publicly available data, therefore a Maximum Likelihood calculation falls apart. Therefore, by implementing MAP instead of model fitting into known potentially right

answers, the model aims to learn the most possible configuration of the network given the structural priors (eg latent tiers $z_i$) and regulatory rules.

- Real World Evaluation: Evaluation on real data has to rely on indirect evidence: temporal stability of the recovered $W_t$ sequence, and comparison against a classical baseline (§8).

## 1.8 7. Design rationale

- **SBM over preferential-attachment/fitness models** as the sole structural backbone: core-periphery tiering in interbank markets is literally a latent-block story, so SBM is the natural fit. PA/fitness models (Caldarelli et al.) are cited in related work for contrast but not built.
- **GRU, not a temporal-GNN** (EvolveGCN, TGN, etc.) for the temporal mechanism: deliberately the simpler choice, given the "big model, small data" risk — roughly 50–80 institutions, at most ~170 periods. A fancier temporal-GNN would be over-parameterized for this panel size.
- **Mandatory baseline**: head-to-head comparison against Gandy–Veraart / RAS maximum-entropy network reconstruction, on both the synthetic SBM data (known ground truth) and the real panel. Explicit instruction from the design discussion: report honestly if the deep model doesn't beat the classical baseline — don't bury a negative result.

## 1.9 8. Variable inventory (what actually feeds the model)

Net statement from the design discussion: **the reconstruction model needs ~5 covariates per node per period, plus the two interbank marginals. Everything else in the NRB bulletins is either a Paper 2 outcome variable or not usable given current data coverage.**

### 1.9.1 8a. The two marginals (non-negotiable — the reconstruction target)

The names below (`2.b Interbank Borrowing`, `INTERBANK LENDING (Marginal r_i)`) were this doc's own shorthand, not literal bulletin text, and the row-marginal name additionally implied a standalone top-level item that doesn't exist. Confirmed against the real extracted CSV (`nrb_balance_sheet.csv`) on 2026-10-22 — see §9.4:

| Symbol | Real bulletin text | Parent section | Statement / side |
|---|---|---|---|
| $c_{i,t}$ (column marginal) | `b. Interbank Borrowing` | `2 BORROWINGS` | Balance Sheet, **Liabilities** |
| $r_{i,t}$ (row marginal) | `a. Interbank Lending` | `3 SHARE & OTHER INVESTMENT` | Balance Sheet, **Assets** |

Both are lettered sub-items, not top-level rows — `r_{i,t}$` in particular is a breakdown component of `3 SHARE & OTHER INVESTMENT`'s own reported total, which matters for the size formula below (§9.4). These are the actual payoff variables for the whole method — everything else exists to help infer the matrix whose row/column sums these fix.

### 1.9.2 8b. Node covariates $X_{i,t}$ (block-membership / fitness prior)

| # | Covariate | Built from | Role |
|---|---|---|---|
| 1 | Size | sum of the **10** real top-level asset rows: `1 LIQUID FUNDS`, `2 INVESTMENT IN SECURITIES`, `3 SHARE & OTHER INVESTMENT`, `4 LOANS & ADVANCES`, `5 BILL PURCHASED`, `6 LOANS AGAINST COLLECTED BILLS`, `7 FIXED ASSETS`, `8 OTHER ASSETS`, `9 EXPENSES NOT WRITTEN OFF`, `10 NON BANKING ASSETS` (**not** the `TOTAL ASSETS` label directly — see §9.1). $r_{i,t}$ is **not** added as an 11th term — per §8a/§9.4 it's a sub-item already folded into `3 SHARE & OTHER INVESTMENT`'s reported total, so adding it separately would double-count it | SBM block/fitness prior, scale control |
| 2 | Capital ratio | `1 CAPITAL FUND` (Liabilities) ÷ size | solvency/fitness term; also sets the obligor cap $K_{i,t}$ |
| 3 | Liquidity ratio | `1 LIQUID FUNDS` (Assets) ÷ size | liquidity position |
| 4 | Loan-to-deposit pressure | `4 LOANS & ADVANCES` (Assets) ÷ `3 DEPOSITS` (Liabilities) | funding-pressure signal |

| # | Covariate | Built from | Role |
|---|---|---|---|
| 5 | NRB-dependence | `a. NRB` ÷ `2 BORROWINGS` (both under Liabilities, `a. NRB` nested under `2 BORROWINGS`; if `2 BORROWINGS` is missing for a row — some early-era filings never print the top-level subtotal — the denominator is reconstructed as the sum of its sub-items: `a. NRB`, `b. Interbank Borrowing`, `c. Foreign Banks and Fin. Ins.`, `d. Other Financial Ins.`, `e. Bonds and Securities` — confirmed real text per §9.4; the `f./g.` sub-items this doc previously assumed existed under `2 BORROWINGS` have **not** been confirmed in the real data yet) | central-bank liquidity dependence — useful for the zero-inflation gate (a bank funded mainly by NRB is less likely to have interbank exposure) |

Optional secondary signal, not in the core 5: ownership type (state/private/foreign), if available — would add homophily structure to the block prior.

#### 1.9.3 8c. Auxiliary observed-graph candidate (optional GNN refinement input, §4)

| Source | Role |
|---|---|
| Statement of Credit, sector-wise breakdown (12 sectors + Others) | shared sector-concentration = a correlated-shock signal independent of the interbank marginals. Candidate as the auxiliary *observed* graph the GNN layer can refine over, since it doesn't require $W_t$ itself. |

#### 1.9.4 8d. Explicitly excluded from Paper 1

- `6 Bills Payable`, `5 Other Liabilities` (Liabilities, minor residuals) — not used
- `6/10 Reconciliation A/c` (both sides) — accounting plug, not economically meaningful
- `8 Expenses not Written off`, `9 Non Banking Assets` (Assets) — legacy/minor, not used
- Everything in Profit & Loss — these are Paper 2 outcome variables, not reconstruction inputs (see `paper2_causal_spillover.md`)
- `nrb_statement_of_credit.csv` as a panel source — only 2 years of coverage in current data, can't support a 14-year panel; not worth cleaning further unless more years turn up uncollected

#### 1.9.5 8e. One-to-one equation-symbol raw-variable map

Every symbol that appears in Eq. (1)–(8) is listed below with exactly what real NRB field (if any) it is computed from. Symbols with no raw field are latent or learned — they're estimated by the model, not read off the bulletin.

| Symbol | Appears in | Raw NRB field | Statement / side | Transformation |
|---|---|---|---|---|
| $r_{i,t}$ | Eq. (5), (8) | `a. Interbank Lending` (sub-item of `3 SHARE & OTHER INVESTMENT`) | Balance Sheet, Assets | identity (reported value) |
| $c_{i,t}$ | Eq. (5), (8) | `b. Interbank Borrowing` (sub-item of `2 BORROWINGS`) | Balance Sheet, Liabilities | identity (reported value) |
| $K_{i,t}$ | Eq. (8), $\kappa K_{i,t}$ cap | `1 CAPITAL FUND` | Balance Sheet, Liabilities | identity (reported value) — note: raw value here, *not* the ratio used in $X^{(2)}_{i,t}$ below |

| Symbol | Appears in | Raw NRB field | Statement / side | Transformation |
|---|---|---|---|---|
| $X_{i,t}^{(1)}$ | Eq. (2), component of $X_{i,t}$ | sum of the 10 real top-level asset rows (listed in §8b) | Balance Sheet, Assets | sum (not the `TOTAL ASSETS` label — see §9.1; $r_{i,t}$ excluded, see §8b/§9.3) |
| $X_{i,t}^{(2)}$ | Eq. (2) | `1 CAPITAL FUND` | Balance Sheet, Liabilities | $\div X_{i,t}^{(1)}$ (capital ratio) |
| $X_{i,t}^{(3)}$ | Eq. (2) | `1 LIQUID FUNDS` | Balance Sheet, Assets | $\div X_{i,t}^{(1)}$ (liquidity ratio) |
| $X_{i,t}^{(4)}$ | Eq. (2) | `4 LOANS & ADVANCES ÷ 3 DEPOSITS` | Balance Sheet, Assets ÷ Liabilities | ratio across statement sides |
| $X_{i,t}^{(5)}$ | Eq. (2) | `a. NRB ÷ 2 BORROWINGS` | Balance Sheet, Liabilities (both) | ratio within Liabilities; denominator falls back to the sum of `a.`–`e.` sub-items (§8b) when `2 BORROWINGS` is unreported |
| $w$ | Eq. (2) | — | — | **learned** (fitness weight vector, dim = 5, no raw field) |
| $z_i$ | Eq. (1), (3), (7) | — | — | **latent** (inferred via $q_\psi$, Eq. 7); optionally informed by ownership type if used (§8b) |
| $\Lambda$ | Eq. (3) | — | — | **learned** ($K \times K$ block-pair intensity matrix) |

| Symbol | Appears in | Raw NRB field | Statement / side | Transformation |
|---|---|---|---|---|
| $h_{i,t}$ | Eq. (3), (6), (7) | $X_{i,t}^{(1\text{-}5)}$ (all 5 covariates, fed in as the GRU input vector) | — | **learned embedding**, GRU output — not read directly off any single field |
| $\theta$ (GRU weights) | Eq. (6) | — | — | **learned** |
| $\psi$ (posterior-net weights) | Eq. (7) | — | — | **learned** |
| $\eta_{ij,t}$ | Eq. (4) | — | — | **learned** (zero-inflation logit, decoder output) |
| $\sigma^2$ | Eq. (4) | — | — | **learned** (log-normal dispersion — global or per-block, §10) |
| $\kappa$ | Eq. (8) | — | — | **exogenous regulatory constant** (single-obligor exposure limit; not reported in the bulletin itself — a known regulatory parameter) |
| $A^{\text{obs}}$ (auxiliary GNN graph, §4) | Eq. (6) refinement, optional | sector-wise credit, 12 sectors | Statement of Credit | co-exposure adjacency built from shared-sector lending shares |

**Reading guide**: $X_{i,t} = (X_{i,t}^{(1)}, ..., X_{i,t}^{(5)}) \in \mathbb{R}^5$ is the concrete instantiation of the generic $X_{i,t} \in \mathbb{R}^d$ in §2 — i.e. $d = 5$ for this model, and the ordering above (size, capital ratio, liquidity ratio, LDR, NRB-dependence) is the fixed component order used everywhere $X_{i,t}$ appears in Eq. (2) and (6).

## 1.10 9. Known data-pipeline caveats (apply before this goes into the model)

1. **`item_code` collision**: confirmed by re-running `nrb_extractor.py` against a real bulletin — `item_code` is not unique across the Balance Sheet's Liabilities and Assets sides. Both sides independently restart numbering at 1 (e.g. `1 CAPITAL FUND` on Liabilities vs. `1 LIQUID FUNDS` on Assets; same collision pattern for `2`, `3`, and lettered sub-items `a.`–`g.`). The extractor (`nrb_extractor.py:197-269`, `_extract_rows`) currently has no column marking which side a row is on — that currently has to be inferred externally from which `TOTAL` block (`TOTAL Liabilities` vs. `TOTAL ASSETS`) a row falls between on the page. **Fix needed**: add a `side` column to the extractor output before any join on `item_code` alone, or it will silently collide liabilities and assets rows that share a code (e.g. covariates 2 and 3 above, which both use an `item_code == 1` row from different sides).
2. Rotated bank-name header recovery (`recover_bank_names`, `nrb_extractor.py:130-171`) and the layout-mode row extraction are both working as validated against the real sample PDF — no open bug there as of this pass.
3. **`item_code` drift across eras**: the top-level code attached to a given labeled row is not stable over the panel's history — e.g. `LOANS & ADVANCES` is coded `4` in some eras and `4.1` in others (with `BILL PURCHASED` shifting from `5` to `4.2` correspondingly). Matching must key off the normalized label text (and, for lettered sub-items, parent section), never off `item_code` alone.
4. **Lettered sub-items (`a.`–`g.`) are reused across unrelated parent sections** (e.g. `a.` means `Paid-up Capital` under `1 CAPITAL FUND`, `Cash Balance` under `1 LIQUID FUNDS`, `NRB` under `2 BORROWINGS`, and `Interbank Lending` under `3 SHARE & OTHER INVESTMENT` — all in the same bulletin). The letter alone is not a key; a row's identity is (normalized label text, nearest preceding top-level/bare-number row in extraction order). The extractor should carry this parent link explicitly (a `parent_item` or `section` column) rather than requiring every consumer to replay the state machine itself.
5. **Truncated-text duplicate rows**: some extractions clip a label mid-word (e.g. `4 LOANS &` alongside the correct `4 LOANS & ADVANCES`, or `1 CAPITAL` alongside `1 CAPITAL FUND`) for the same `item_code`/concept. Confirmed via `cut -d, -f7,8 ... | sort -u` on the real CSV (2026-10-22). Treat these as the same field (prefix match or longest-label-wins), not as missing data for the truncated variant.

## 1.11 10. Open items (not yet decided)

- $K$ **(number of SBM blocks)**: start with $K = 2$ (core/periphery), treat larger $K$ as a robustness check, not the headline specification.
- $\sigma^2$ **(log-normal dispersion)**: learned, either globally or per-block — not yet decided which.
- **Venue/scope hook**: ICAIF fit is comfortable at the current scope (methods + real-data case study). ICLR/NeurIPS main track would need an added theoretical hook — e.g. a provable statement on how reconstruction uncertainty in $\hat{W}_t$ propagates into coverage guarantees for a down-

stream interference estimator (which would also strengthen the Paper 1 → Paper 2 link). **Not resolved** — needs a decision before drafting.

### 1.12 11. Algorithms (pseudocode)

Three algorithms, in dependency order: the capacitated Sinkhorn projection (the hard constraint layer, Eq. 8) is standalone and testable without the rest of the model; the generative forward pass (Eq. 1–5, 8) samples a synthetic $W_t$ for validation; training (Eq. 6–9) wraps the amortized encoder/decoder of §4 around the same projection layer. All three are written over *symbol-level* inputs ($N, K, d, \kappa$, etc.) — nothing below is a fixed dataset size or constant; see §12 for how these get threaded through as config, not literals.

**Algorithm 1** CapacitatedSinkhornProjection (Eq. 8)

**Require:** $\tilde{W} \in \mathbb{R}_{\geq 0}^{N \times N}$ (raw/sampled edge weights), $r, c \in \mathbb{R}_{\geq 0}^{N}$ (target marginals), $\text{cap} \in \mathbb{R}_{\geq 0}^{N}$ (per-row cap, $\text{cap}_i = \kappa \cdot K_{i,t}$), $\varepsilon > 0$ (entropic regularization), $n_{\text{iters}}$, tol (stopping criteria)
**Ensure:** $W \in \mathbb{R}_{\geq 0}^{N \times N}$ satisfying row/col marginals and $W_{ij} \leq \text{cap}_i$ exactly

$\text{Kern} \leftarrow \exp(\log(\tilde{W} + \delta)/\varepsilon)$ ▷ $\delta$ tiny avoids log(0); config, not literal
$u \leftarrow \mathbf{1}_N$; $v \leftarrow \mathbf{1}_N$
**for** iter $= 1 \ldots n_{\text{iters}}$ **do**
  $u_{\text{new}} \leftarrow r/(\text{Kern} \cdot v)$
  $u_{\text{new}} \leftarrow \min(u_{\text{new}}, \text{cap}/(\text{Kern} \cdot v))$ ▷ Enforce row-wise cap
  $v_{\text{new}} \leftarrow c/(\text{Kern}^T \cdot u_{\text{new}})$
  **if** $\|u_{\text{new}} - u\| < \text{tol}$ **and** $\|v_{\text{new}} - v\| < \text{tol}$ **then**
    **break**
  **end if**
  $u, v \leftarrow u_{\text{new}}, v_{\text{new}}$
**end for**
$W \leftarrow \text{diag}(u) \cdot \text{Kern} \cdot \text{diag}(v)$
**return** $W$

**Algorithm 2** SampleSyntheticNetwork (Eq. 1–5, 8; ground truth for validation, §6)

**Require:** $N, K, d$ (problem dimensions), $\pi \in \Delta^K, \Lambda \in \mathbb{R}_{\geq 0}^{K \times K}, w \in \mathbb{R}^d, \sigma^2$ (generative params), $X_{1:N} \in \mathbb{R}^{N \times d}$ (covariates), $\kappa, K_{\text{cap}} \in \mathbb{R}^N$ (regulatory cap inputs)
**Ensure:** $W_{\text{true}} \in \mathbb{R}_{\geq 0}^{N \times N}, r, c \in \mathbb{R}^N$ (ground-truth network + marginals)

**for** $i = 1 \ldots N$ **do**
  $z_i \sim \text{Categorical}(\pi)$ ▷ Eq. 1
**end for**
**for** $i = 1 \ldots N$ **do**
  $\varphi_i \leftarrow \exp(w^T X_i)$ ▷ Eq. 2
**end for**
**for** $i, j = 1 \ldots N$ $(i \neq j)$ **do**
  $\lambda_{ij} \leftarrow \Lambda[z_i, z_j] \cdot \varphi_i \cdot \varphi_j$ ▷ Eq. 3
  $e_{ij} \sim \text{Bernoulli}(\sigma(\eta_{ij}))$ ▷ Eq. 4, zero-inflation param
  $\tilde{W}_{ij} \leftarrow e_{ij} \cdot \text{LogNormalSample}(\log \lambda_{ij}, \sigma^2)$
**end for**
$r \leftarrow \text{row_sums}(\tilde{W})$; $c \leftarrow \text{col_sums}(\tilde{W})$
$W_{\text{true}} \leftarrow \text{CapacitatedSinkhornProjection}(\tilde{W}, r, c, \kappa \cdot K_{\text{cap}})$ ▷ Eq. 8
**return** $W_{\text{true}}, \text{row_sums}(W_{\text{true}}), \text{col_sums}(W_{\text{true}})$

Sinkhorn (operating on a non-negative matrix, enforcing row/col marginals  0 via the kernel) is only the last step. The full pipeline per Algorithm 3 (§11) and Eq. 1-8:

**Algorithm 3** TrainStep (Eq. 6–9; amortized inference + one optimizer step)

**Require:** Batch of periods $\{X_{1:N,t}, r_t, c_t, K_t\}_{t\in\text{batch}}$, optional $W_{\text{true},t}$, model params $\theta, \psi, \Lambda, w, \sigma^2$, hyperparams $\lambda_{\text{smooth}}, \lambda_{\text{reg}}, \kappa$
**State:** $h_{i,t-1}$ carried across periods (GRU hidden state)
**Ensure:** Loss value; updated $\theta, \psi, \Lambda, w$ after one gradient step

**for** $t \in$ batch (temporal order) **do**
  **for** $i = 1 \dots N$ **do**
    $h_{i,t} \leftarrow \text{GRU}_\theta(X_{i,t}, h_{i,t-1})$ ▷ Eq. 6
    [Optional] $h_{i,t} \leftarrow \text{GNN_refine}(h_{i,t}, A_{\text{obs},t})$ ▷ §4, §8c
    $q_i \leftarrow \text{softmax}(\text{MLP}_\psi(h_{i,t}))$ ▷ Eq. 7
    $z_i \sim \text{GumbelSoftmaxSample}(q_i)$ ▷ Reparameterized
  **end for**
  **for** $i, j = 1 \dots N$ $(i \neq j)$ **do**
    $(\eta_{ij,t}, \lambda_{ij,t}) \leftarrow \text{Decoder}_\psi(h_{i,t}, h_{j,t}, z_i, z_j)$ ▷ Eq. 3–4
    $e_{ij,t} \sim \text{GumbelSigmoidSample}(\eta_{ij,t})$ ▷ Reparameterized gate
    $\tilde{W}_{ij,t} \leftarrow e_{ij,t} \cdot \text{LogNormalSample}(\log \lambda_{ij,t}, \sigma^2)$
  **end for**
  $W_t \leftarrow \text{CapacitatedSinkhornProjection}(\tilde{W}_t, r_t, c_t, \kappa \cdot K_t)$ ▷ Eq. 8, Alg. 1
  **if** $W_{\text{true},t}$ is given (synthetic branch) **then**
    $L_{\text{recon},t} \leftarrow -\log p(W_{\text{true},t} \mid \lambda_t, \eta_t, \sigma^2)$ ▷ Eq. 9, first term
  **else**
    $L_{\text{recon},t} \leftarrow 0$ ▷ Real data: no ground truth
  **end if**
  $L_{\text{KL},t} \leftarrow \text{KL}(q_\psi(z_t) \,\|\, p(z))$
**end for**
$L_{\text{smooth}} \leftarrow \sum_t \|W_t - W_{t-1}\|_F^2$ ▷ Over consecutive periods in batch
$L_{\text{reg}} \leftarrow \|\Lambda\|^2 + \|w\|^2$
$L \leftarrow \sum_t (L_{\text{recon},t} + L_{\text{KL},t}) + \lambda_{\text{smooth}} \cdot L_{\text{smooth}} + \lambda_{\text{reg}} \cdot L_{\text{reg}}$
Backpropagate $L$; update $\theta, \psi, \Lambda, w$ via optimizer step
**return** $L$

- 1. SBM block assignment + covariate fitness → gives each bank pair a prior edge intensity $\lambda_{ij,t}$ (Eq. 1-3), informed by $z_i$ and the 5 covariates.
- 2. GNN/GRU encoder + pairwise decoder → turns that into $(\eta_{ij,t}, \lambda_{ij,t})$ per pair, learned from data (Eq. 6-7, §4).
- 3. Zero-inflated log-normal sampling → produces the raw guess $\tilde{W}_t$ (Eq. 4) — this is where the actual "who's connected to whom, roughly how much" signal comes from.
- 4. Only then does Sinkhorn project $\tilde{W}_t$ onto the feasible set that exactly matches the observed marginals $r_t, c_t$ and the regulatory cap (Eq. 8) — a correction/projection step, not a generative one.

In summary, plain Sinkhorn/IPF applied to a uniform prior (no covariates, no learned structure) is exactly the classical Gandy-Veraart/RAS maximum-entropy reconstruction — that's the baseline this paper compares against (§7, §12 baseline requirement). The contribution is that Sinkhorn here operates on a covariate- and block-structure-informed $\tilde{W}_t$, not blind uniform mass — so you justify why Sinkhorn (it's the natural, differentiable way to enforce known hard constraints exactly, à la Cuturi 2013) separately from why not Sinkhorn alone (because unlike RAS/Gandy-Veraart, the prior it's projecting isn't uninformed).

## Methodological Literature Synthesis

The model bridges three major streams of literature: classical financial network reconstruction, entropic optimal transport, and amortized deep latent variable modeling.

### Classical Interbank Matrix Reconstruction

Early efforts to estimate missing bilateral exposures ($W_t$) from aggregated call reports relied primarily on Maximum Entropy (ME) principles [**upper2011simulation**]. While ME models successfully match row and column balance sheet totals ($r_t, c_t$), they typically yield fully connected, overly dense matrices that understate systemic fragility. Modern maximum-entropy and fitness framework extensions attempt to introduce sparsity [**squartini2018reconstruction**], yet these classical models rely on rigid functional forms and cannot easily learn complex temporal dynamics or non-linear node covariate signals ($X_{i,t}$).

### Entropic Optimal Transport & Differentiable Constraints

To enforce macro-prudential constraints differentiably, the model builds on entropic optimal transport (EOT) frameworks. [**cuturi2013sinkhorn**] demonstrated that adding an entropic regularizer to optimal transport transforms non-differentiable linear programs into computationally efficient, GPU-parallelizable scaling operations via the Sinkhorn-Knopp algorithm. Further extensions by [**mena2018learning**] showed that applying the Sinkhorn operator directly to neural network outputs enables gradient backpropagation through continuous matrix transformations and discrete topology

decisions. In our pipeline, this capacitated Sinkhorn projection enforces row/column totals $(r_t, c_t)$ and regulatory single-obligor exposure limits $(W_{ij,t} \leq \kappa K_{i,t})$ directly inside the forward pass.

### Amortized Variational Inference for Graphs

Standard Graph Autoencoders require observing the graph adjacency matrix to compute node representations. To bypass this circularity on unobserved networks, our inference architecture draws inspiration from Neural Relational Inference (NRI) [**kipf2018neural**]. NRI established that latent relational structures can be inferred purely from dynamic node trajectories $(X_{1:T})$ using a graph variational autoencoder. By conditioning the GNN/GRU temporal encoder strictly on independent node covariates and history rather than unobserved edges, our pipeline avoids circular identification logic while predicting continuous weight distributions and discrete connection gates.

## 1.13 8. Variable inventory (what actually feeds the model)

Net statement from the design discussion: **the reconstruction model needs ~5 covariates per node per period, plus the two interbank marginals. Everything else in the NRB bulletins is either a Paper 2 outcome variable or not usable given current data coverage.**

### 1.13.1 8a. The two marginals (non-negotiable — the reconstruction target)

| NRB field | Statement / side | Symbol |
|---|---|---|
| `2.b Interbank Borrowing` | Balance Sheet, **Liabilities** | $c_{i,t}$ (column marginal) |
| `INTERBANK LENDING (Marginal r_i)` | Balance Sheet, **Assets** | $r_{i,t}$ (row marginal) |

These are the actual payoff variables for the whole method — everything else exists to help infer the matrix whose row/column sums these fix.

### 1.13.2 8b. Node covariates $X_{i,t}$ (block-membership / fitness prior)

| # | Covariate | Built from | Role |
|---|---|---|---|
| 1 | Size | sum of the 11 `ASSET_TOP_LEVEL` columns: `1 LIQUID FUNDS`, `2 INVESTMENT IN SECURITIES`, `3 SHARE & OTHER INVESTMENT`, `4 LOANS & ADVANCES`, `5 BILL PURCHASED`, `6 LOANS AGAINST COLLECTED BILLS`, `7 FIXED ASSETS`, `8 OTHER ASSETS`, `9 EXPENSES NOT WRITTEN OFF`, `10 NON BANKING ASSETS`, `INTERBANK LENDING (Marginal r_i)` (**not** the `TOTAL ASSETS` label directly — see data caveat below) | SBM block/fitness prior, scale control |
| 2 | Capital ratio | `1 CAPITAL FUND` (Liabilities) ÷ size | solvency/fitness term; also sets the obligor cap $K_{i,t}$ |
| 3 | Liquidity ratio | `1 LIQUID FUNDS` (Assets) ÷ size | liquidity position |
| 4 | Loan-to-deposit pressure | `4 LOANS & ADVANCES` (Assets) ÷ `3 DEPOSITS` (Liabilities) | funding-pressure signal |

| # | Covariate | Built from | Role |
|---|---|---|---|
| 5 | NRB-dependence | `2.a Borrowings - NRB` ÷ `2 BORROWINGS` (both Liabilities; if `2 BORROWINGS` is missing for a row — some early-era filings never print the top-level subtotal — the denominator is reconstructed as the sum of its sub-items: `2.a Borrowings - NRB`, `2.b Interbank Borrowing`, `2.c Foreign Banks and Financial Institutions`, `2.d Other Financial Institutions`, `2.e Bonds and Securities / NRB Bonds`, `2.f Borrowings - Govt Non-Financial Institutions`, `2.g Borrowings - Other Non-Financial Institutions`) | central-bank liquidity dependence — useful for the zero-inflation gate (a bank funded mainly by NRB is less likely to have interbank exposure) |

Optional secondary signal, not in the core 5: ownership type (state/private/foreign), if available — would add homophily structure to the block prior.

#### 1.13.3 8c. Auxiliary observed-graph candidate (optional GNN refinement input, §4)

| Source | Role |
|---|---|
| Statement of Credit, sector-wise breakdown (12 sectors + Others) | shared sector-concentration = a correlated-shock signal independent of the interbank marginals. Candidate as the auxiliary *observed* graph the GNN layer can refine over, since it doesn't require $W_t$ itself. |

#### 1.13.4 8d. Explicitly excluded from Paper 1

- `6 Bills Payable`, `5 Other Liabilities` (Liabilities, minor residuals) — not used
- `6/10 Reconciliation A/c` (both sides) — accounting plug, not economically meaningful
- `8 Expenses not Written off`, `9 Non Banking Assets` (Assets) — legacy/minor, not used
- Everything in Profit & Loss — these are Paper 2 outcome variables, not reconstruction inputs (see `paper2_causal_spillover.md`)
- `nrb_statement_of_credit.csv` as a panel source — only 2 years of coverage in current data, can't support a 14-year panel; not worth cleaning further unless more years turn up uncollected

#### 1.13.5 8e. One-to-one equation-symbol raw-variable map

Every symbol that appears in Eq. (1)–(8) is listed below with exactly what real NRB field (if any) it is computed from. Symbols with no raw field are latent or learned — they're estimated by the model, not read off the bulletin.

| Symbol | Appears in | Raw NRB field | Statement / side | Transformation |
|---|---|---|---|---|
| $r_{i,t}$ | Eq. (5), (8) | `INTERBANK LENDING (Marginal r_i)` | Balance Sheet, Assets | identity (reported value) |
| $c_{i,t}$ | Eq. (5), (8) | `2.b Interbank Borrowing` | Balance Sheet, Liabilities | identity (reported value) |
| $K_{i,t}$ | Eq. (8), $\kappa K_{i,t}$ cap | `1 CAPITAL FUND` | Balance Sheet, Liabilities | identity (reported value) — note: raw value here, *not* the ratio used in $X^{(2)}_{i,t}$ below |
| $X^{(1)}_{i,t}$ | Eq. (2), component of $X_{i,t}$ | sum of the 11 `ASSET_TOP_LEVEL` columns (listed in §8b) | Balance Sheet, Assets | sum (not the `TOTAL ASSETS` label — see §9.1) |

| Symbol | Appears in | Raw NRB field | Statement / side | Transformation |
|---|---|---|---|---|
| $X^{(2)}_{i,t}$ | Eq. (2) | `1 CAPITAL FUND` | Balance Sheet, Liabilities | $\div X^{(1)}_{i,t}$ (capital ratio) |
| $X^{(3)}_{i,t}$ | Eq. (2) | `1 LIQUID FUNDS` | Balance Sheet, Assets | $\div X^{(1)}_{i,t}$ (liquidity ratio) |
| $X^{(4)}_{i,t}$ | Eq. (2) | `4 LOANS & ADVANCES ÷ 3 DEPOSITS` | Balance Sheet, Assets ÷ Liabilities | ratio across statement sides |
| $X^{(5)}_{i,t}$ | Eq. (2) | `2.a Borrowings - NRB ÷ 2 BORROWINGS` | Balance Sheet, Liabilities (both) | ratio within Liabilities; denominator falls back to the sum of `2.a`–`2.g` sub-items when `2 BORROWINGS` is unreported (§8b) |
| $w$ | Eq. (2) | — | — | **learned** (fitness weight vector, dim = 5, no raw field) |
| $z_i$ | Eq. (1), (3), (7) | — | — | **latent** (inferred via $q_\psi$, Eq. 7); optionally informed by ownership type if used (§8b) |
| $\Lambda$ | Eq. (3) | — | — | **learned** ($K \times K$ block-pair intensity matrix) |
| $h_{i,t}$ | Eq. (3), (6), (7) | $X^{(1\text{-}5)}_{i,t}$ (all 5 covariates, fed in as the GRU input vector) | — | **learned embedding**, GRU output — not read directly off any single field |
| $\theta$ (GRU weights) | Eq. (6) | — | — | **learned** |
| $\psi$ (posterior-net weights) | Eq. (7) | — | — | **learned** |

| Symbol | Appears in | Raw NRB field | Statement / side | Transformation |
|---|---|---|---|---|
| $\eta_{ij,t}$ | Eq. (4) | — | — | **learned** (zero-inflation logit, decoder output) |
| $\sigma^2$ | Eq. (4) | — | — | **learned** (log-normal dispersion — global or per-block, §10) |
| $\kappa$ | Eq. (8) | — | — | **exogenous regulatory constant** (single-obligor exposure limit; not reported in the bulletin itself — a known regulatory parameter) |
| $A^{\text{obs}}$ (auxiliary GNN graph, §4) | Eq. (6) refinement, optional | sector-wise credit, 12 sectors | Statement of Credit | co-exposure adjacency built from shared-sector lending shares |

**Reading guide**: $X_{i,t} = (X_{i,t}^{(1)}, ..., X_{i,t}^{(5)}) \in \mathbb{R}^5$ is the concrete instantiation of the generic $X_{i,t} \in \mathbb{R}^d$ in §2 — i.e. $d = 5$ for this model, and the ordering above (size, capital ratio, liquidity ratio, LDR, NRB-dependence) is the fixed component order used everywhere $X_{i,t}$ appears in Eq. (2) and (6).

### 1.14 9. Known data-pipeline caveats (apply before this goes into the model)

1. **`item_code` collision**: confirmed by re-running `nrb_extractor.py` against a real bulletin — `item_code` is not unique across the Balance Sheet's Liabilities and Assets sides. Both sides independently restart numbering at 1 (e.g. `1 CAPITAL FUND` on Liabilities vs. `1 LIQUID FUNDS` on Assets; same collision pattern for `2`, `3`, and lettered sub-items `a.`–`g.`). The extractor (`nrb_extractor.py:197-269`, `_extract_rows`) currently has no column marking which side a row is on — that currently has to be inferred externally from which `TOTAL` block (`TOTAL Liabilities` vs. `TOTAL ASSETS`) a row falls between on the page. **Fix needed**: add a `side` column to the extractor output before any join on `item_code` alone, or it will silently collide

liabilities and assets rows that share a code (e.g. covariates 2 and 3 above, which both use an `item_code == 1` row from different sides).

2. Rotated bank-name header recovery (`recover_bank_names`, `nrb_extractor.py:130-171`) and the layout-mode row extraction are both working as validated against the real sample PDF — no open bug there as of this pass.

### 1.15 10. Open items (not yet decided)

- $K$ (number of SBM blocks): start with $K = 2$ (core/periphery), treat larger $K$ as a robustness check, not the headline specification.
- $\sigma^2$ (log-normal dispersion): learned, either globally or per-block — not yet decided which.
- **Venue/scope hook**: ICAIF fit is comfortable at the current scope (methods + real-data case study). ICLR/NeurIPS main track would need an added theoretical hook — e.g. a provable statement on how reconstruction uncertainty in $\hat{W}_t$ propagates into coverage guarantees for a downstream interference estimator (which would also strengthen the Paper 1 → Paper 2 link). **Not resolved** — needs a decision before drafting.

### 1.16 11. Paper structure ( 7 pages)

| Section | Length | Content |
|---|---|---|
| Intro + related work | ~1.5p | position against Aronow-Samii/Leung (network assumed *known*), Elliott-Golub-Jackson (network assumed known, no reconstruction), Gandy-Veraart (reconstruction without covariate-informed priors or regulatory constraints) |
| Method | ~2.5p | §3–6 above |
| Synthetic validation | ~1.5p | SBM-generated ground truth, baseline comparison |
| Real-data case study | ~1p | Nepal panel, temporal-stability evidence |
| Discussion/limitations | ~0.5p | honest treatment of the no-ground-truth-on-real-data limitation |